

§ 9. Extension of the investigation to modules. Equality of the number of components in decompositions into irreducible modules.

We now show that the content of the first three paragraphs, which concerns irreducible ideals and not primary and prime ideals, remains valid under weaker hypotheses. Those paragraphs do not use the commutative law of multiplication and refer only to the property of ideals of being modules. They therefore remain valid for modules with respect to non-commutative domains, which are now to be defined. The definition of modules is to be based on a double domain (Σ, T) with the following properties.

Σ is an abstractly defined non-commutative ring, that is, Σ is a system of elements $a, b, c, \dots$ for which two modes of combination are defined, ring addition (+) and ring multiplication ($\times$), satisfying the laws stated in § 1, with the exception of the commutative law 4 for ring multiplication.

T is a system of elements $\alpha, \beta, \gamma, \dots$, for which, in connection with Σ , two combinations are likewise defined: addition, which from any two elements α, β produces a unique third element $\alpha + \beta$; and multiplication of an element α of T by an element c of Σ , which uniquely produces an element $c\alpha$ of T .³⁵

For these combinations the following laws hold:

1. The associative law of addition: $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.
2. The commutative law of addition: $\alpha + \beta = \beta + \alpha$.
3. The law of unrestricted and unique subtraction: there is in T one and only one element ξ satisfying the equation $\alpha + \xi = \beta$ (one writes $\xi = \beta - \alpha$).
4. The associative law of multiplication: $a(b\gamma) = (a \times b)\gamma$.
5. The distributive law:

$$(a + b)\gamma = a\gamma + b\gamma, \quad c(\alpha + \beta) = c\alpha + c\beta.$$

From these conditions there follow, as is well known, the existence of the zero element and the validity of the distributive law also for the combination of subtraction and multiplication:

$$(a - b)\gamma = a\gamma - b\gamma, \quad c(\alpha - \beta) = c\alpha - c\beta,$$

where $(-)$ denotes subtraction in Σ . If Σ contains a unit ε , then $\varepsilon\alpha = \alpha$ is to hold for every element α of T .

By a module M in (Σ, T) we mean a system of elements of T satisfying the two conditions:

1. along with α , M contains $c\alpha$, where c is any element of Σ ;
2. along with α and β , M contains the difference $\alpha - \beta$, hence also $n\alpha$ for every integer n .³⁶

By this definition, T itself forms a module in (Σ, T) . If in particular the domain T and the combinations established there coincide with the domain Σ and the combinations valid there, then the module M becomes a right ideal M in Σ . If Σ is also assumed commutative, the ordinary concept of ideal results, and is thus a special case of the concept of module.³⁷

For modules all definitions of § 1 remain valid. Thus $\alpha \equiv 0(M)$, respectively $N \equiv 0(M)$, means that α , respectively every element of N , is an element of M ; in other words, α , respectively N , is divisible by M . M is a proper divisor of N if M contains elements different from those of N ; from $N \equiv 0(M)$ and $M \equiv 0(N)$ follows $M = N$. The definitions of greatest common divisor and least common multiple likewise remain literally the same. If the module M contains, in particular, a finite number of elements $\alpha_1, \dots, \alpha_\rho$ such that

$$M = (\alpha_1, \dots, \alpha_\rho),$$

that is,

$$\alpha = c_1\alpha_1 + \dots + c_\rho\alpha_\rho + n_1\alpha_1 + \dots + n_\rho\alpha_\rho$$

³⁵Thus here one is dealing with a "right-sided" multiplication, a "right-sided" domain T , and consequently with "right-sided" modules and ideals. If one were to base T instead on a left-sided multiplication αc , a corresponding theory of left-sided modules and ideals would result; M would contain αc along with α . The associative law would then have the form $(\gamma b)a = \gamma(b \times a)$.

³⁶The integers are again to be understood as abbreviating symbols, not as ring elements.

³⁷The simplest example of a module is the module of integral linear forms: here Σ consists of all rational integers and T of all integral linear forms. A somewhat more general module arises when, in Σ and T , one takes algebraic integers instead of rational integers, or for instance all even integers. If, instead of the linear forms, one considers each complex of coefficients as an element, then the combinations in Σ and T are in fact different. Ideals in non-commutative polynomial ring domains are the subject of the joint paper of Noether-Schmeidler. Ideals in other special non-commutative domains are treated in Hurwitz's lectures on the number theory of quaternions (Berlin, Springer 1919) and in the works of Du Pasquier cited there.

for every $\alpha \equiv 0(M)$, where the c_i are elements of Σ and the n_i are integers, then M is called a finite module and $\alpha_1, \dots, \alpha_\rho$ a module basis.

In what follows we assume, just as in § 1, only those domains (Σ, T) that satisfy the finiteness condition: every module in (Σ, T) is finite, and hence has a module basis.

For such a domain (Σ, T) , Theorem I on finite chains holds also for modules, as the proof given there shows, and therefore all hypotheses for §§ 2 and 3 are satisfied. Definition I and Lemma I on shortest and reduced representations carry over directly; likewise Theorem II on the representability of every module as the least common multiple of finitely many irreducible modules, where Lemma II shows that every shortest representation of this kind is at the same time reduced. Theorem III also remains valid, expressing reducibility of a module through properties of its complement; from it follows Lemma III, and finally Theorem IV, which asserts equality of the number of components in two different shortest representations of a module as the least common multiple of irreducible modules. Theorem IV also gives Lemma IV as the converse of Lemma I on reduced representations.

Addendum. The same argument shows that all these theorems and definitions remain valid if, for two-sided domains T , that is, domains which are both right- and left-sided, one everywhere understands by a module a two-sided module: one which, along with α , also contains $c\alpha$ and αc , and along with α and β also contains $\alpha - \beta$.

While all these theorems rest only on the concept of divisibility and of least common multiple, the further uniqueness theorems rest essentially on the product concept and therefore do not admit a direct transfer. Thus the definition of a primary ideal and of a prime ideal cannot be transferred to modules, since the product of two elements of T is not defined. The formally possible transfer to non-commutative rings loses its meaning, since here the existence of the associated prime ideal cannot be proved³⁸ and the proof that an irreducible ideal is primary also fails. These two facts form the basis of the following uniqueness theorems. By contrast, if the non-commutative ring has a unit, coprime and coprime-irreducible ideals can be defined, and the arguments used in the proof of Theorem II show that every ideal can be represented as the least common multiple of finitely many pairwise coprime and coprime-irreducible ideals.³⁹

Finally, we mention a sufficient criterion for the finiteness condition to hold in (Σ, T) : if Σ contains a unit and satisfies the finiteness condition, and if T itself is a finite module in (Σ, T) , then every module in (Σ, T) is finite.

Indeed, from the existence of the unit in Σ , for

$$\mathfrak{M} = (f_1, \dots, f_e), \quad f = \bar{b}_1 f_1 + \dots + \bar{b}_e f_e + n_1 f_1 + \dots + n_e f_e,$$

where the $\bar{b}_i$ are elements of Σ and the n_i are integers, there follows also a representation

$$f = b_1 f_1 + \dots + b_e f_e$$

with the b_i in Σ . For since $f_i = \varepsilon f_i$, one has $n_i f_i = n_i \varepsilon f_i$, and since $n_i \varepsilon = (\varepsilon + \dots + \varepsilon)$ belongs to Σ , $b_i = \bar{b}_i + n_i \varepsilon$ is an element of Σ . The hypothesis on T says that every element α of T admits a representation

$$\alpha = \bar{a}_1 \tau_1 + \dots + \bar{a}_k \tau_k + n_1 \tau_1 + \dots + n_k \tau_k,$$

and hence

$$\alpha = a_1 \tau_1 + \dots + a_k \tau_k,$$

the second representation following from the first just as above because $\varepsilon \tau_i = \tau_i$.

As α runs through the elements of a module M in (Σ, T) , the coefficients a_k of τ_k run through an ideal $\mathfrak{M}_k$ of Σ ; by the preceding, for every $a_k \equiv 0(\mathfrak{M}_k)$ one has

$$a_k = b_1 a_k^{(1)} + \dots + b_e a_k^{(e)}.$$

Let $\alpha^{(i)}$ be an element of M for which the coefficient of τ_k is $a_k^{(i)}$. Then $\alpha - b_1 \alpha^{(1)} - \dots - b_e \alpha^{(e)}$ is an element belonging to M which depends only on $\tau_1, \dots, \tau_{k-1}$. The procedure can be repeated for the totality of these elements, which no longer contain τ_k and which form a module M' , and the assertion is proved by finitely many repetitions.

³⁸For from $p_1^\rho \equiv 0(\Omega)$ and $p_2^\sigma \equiv 0(\Omega)$ it does not follow here that $(p_1 - p_2)^{\rho+\sigma} \equiv 0(\Omega)$; likewise, from $(ab)^\rho \equiv 0(\Omega)$ it does not follow that $a^\rho b^\rho \equiv 0(\Omega)$. Thus $\mathfrak{P}$ can be proved neither to be an ideal nor to have the property of a prime ideal. For two-sided ideals $\mathfrak{P}$ can indeed be proved to be an ideal, but not a prime ideal.

³⁹For special, "completely reducible" ideals, uniqueness theorems can be set up here too; compare the joint paper Noether-Schmeidler.

§ 10. Special case of the polynomial domain.

1. Let the underlying ring domain Σ consist of all polynomials in $x_1, \dots, x_n$ with arbitrary complex coefficients. By Hilbert's basis theorem (Ann. 36) the finiteness condition holds for it. The issue is the relation of our theorems to the known theorems of elimination theory and module theory.

This relation is established by the following special case of a known theorem of Hilbert:⁴⁰

If f vanishes for every finite system of values of $x_1, \dots, x_n$ that is a zero of all polynomials of a prime ideal $\mathfrak{P}$ (a zero of $\mathfrak{P}$), then f is divisible by $\mathfrak{P}$. In other words, a prime ideal $\mathfrak{P}$ consists of the totality of polynomials which vanish at its zeros.⁴¹

Thus, if a product fg vanishes for all zeros of $\mathfrak{P}$, at least one factor vanishes; the zeros form an irreducible algebraic variety. Conversely, if this definition of irreducible variety is adopted, then the totality of polynomials vanishing on an irreducible variety forms a prime ideal. Prime ideal and irreducible variety therefore correspond one-to-one. Since $\mathfrak{P}' \equiv 0(\mathfrak{Q})$ and $\mathfrak{Q} \equiv 0(\mathfrak{P})$, the zeros of a primary ideal coincide with those of its associated prime ideal.⁴² Therefore the representation of an ideal as the least common multiple of greatest primary ideals gives a resolution of all zeros of the ideal into irreducible varieties; and, as Lasker showed, the converse also holds. The proof of uniqueness of the associated prime ideals thus corresponds here to the fundamental theorem of elimination theory on the unique decomposability of an algebraic variety into irreducible varieties, and for special polynomial domains, where no unique product representation of the polynomials by irreducible polynomials of the domain and consequently no elimination theory exists, it can serve as an equivalent of that theorem of elimination theory.

The irreducible varieties corresponding to the isolated primary ideals are exactly those which occur in the "minimal resolvent";⁴³ for the zeros of every non-isolated greatest primary ideal are at the same time zeros of at least one isolated one, namely one whose associated prime ideal is divisible by that of the non-isolated ideal. The uniqueness of isolated primary ideals therefore gives, in the exponents, new invariant multiplicity numbers. The uniqueness theorems for the resolution of primary ideals into irreducible ones can likewise be regarded as supplements to elimination theory in the sense of multiplicity.

After these remarks the different decompositions can be interpreted in terms of their behavior with respect to algebraic varieties. Pairwise coprime ideals correspond to varieties having no common zero; for mutually relatively prime ideals, no irreducible variety of one ideal is at the same time a common zero; the greatest primary ideals vanish only on irreducible varieties which are all distinct from one another; in the decomposition into irreducible ideals the same irreducible varieties may occur with multiplicity.

It should also be noted that, instead of the general polynomial domain, the domain of all homogeneous forms may also be taken as the basis; for one readily verifies that the general theorems remain valid for the combinations there in force, where addition is defined only for forms of the same dimension.⁴⁴

A simplest example of the four different decompositions - for which the formulas follow below - is given, according to the preceding, by a line and two intersecting lines skew to it, one of the latter containing, with higher multiplicity, a point different from the point of intersection. The decomposition into coprime-irreducible ideals corresponds to the decomposition into the line and the variety skew to it; in the decomposition into relatively-prime irreducible ideals this variety splits into the two lines; the decomposition into greatest primary ideals corresponds to separating off the point of higher multiplicity, while the decomposition into irreducible ideals forces a resolution of that point.

Taking this point as the origin, the running line as the y -axis, the line intersecting it parallel to the x -axis, and the line skew to it parallel to the z -axis, such a configuration is represented, for example, by the following irreducible ideals:⁴⁵

$$\mathfrak{B}_1 = (x - 1, y), \quad \mathfrak{B}_2 = (y - 1, z), \quad \mathfrak{B}_3 = (x, z), \quad \mathfrak{B}_4 = (x^3, y, z), \quad \mathfrak{B}_5 = (x^2, y^2, z).$$

⁴⁰Über die vollen Invariantensysteme. Math. Ann. 42 (1893), § 3, p. 313.

⁴¹As Lasker showed [Math. Ann. 60 (1905), p. 607], this special case can also be proved directly for homogeneous forms; conversely, Hilbert's theorem follows again from it in the homogeneous and inhomogeneous cases. We can express that theorem as follows: if an ideal $\mathfrak{R}$ vanishes at all zeros of M , then a power of $\mathfrak{R}$ is divisible by M . This theorem, and also the special case, holds only if the range of values of the x is algebraically closed; hence it cannot follow from our theorems alone, but must use the existence of roots, for example at the point where one uses that an ideal whose zeros consist only of $x_1 = 0, \dots, x_n = 0$ contains all products of powers of the x from some dimension on. The remaining proof can be somewhat simplified in comparison with Lasker by using our theorems. For Lasker must also appeal to Hilbert's theorem to prove the decomposition of an ideal into greatest primary ideals.

⁴²Macaulay (compare the introduction) takes this property of a primary ideal, namely to possess an irreducible variety, as the definition; Lasker includes only the concept of the manifold of a variety in the definition and otherwise defines abstractly. The primary ideals vanishing only for $x_1 = 0, \dots, x_n = 0$ occupy a special position for Lasker.

⁴³Compare, for example, J. König, Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, Teubner, 1903), p. 235.

⁴⁴That in the ambiguous case inhomogeneous decompositions can exist beside homogeneous ones is shown by the example $(x^3, xy, y^3) = [(x^3, y), (y^3, x)] = [(xy, x^3, y^3, x + y^3), (xy, x^3, y^3, y + x^2)]$.

⁴⁵Of these, the first three are irreducible as prime ideals; $\mathfrak{B}_4$ because it has only the divisors (x^2, y, z) and (x, y, z) ; $\mathfrak{B}_5$ because every divisor contains the polynomial xy .

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x, y, z).$$

The greatest primary ideals are

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^3, y^2, x^2y, z).$$

The relatively-prime irreducible ideals are

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^3, x^2y, xy^2, z).$$

The coprime-irreducible ideals are

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y-1)x^3, (y-1)x^2y, (y-1)xy^2, z).$$

Thus the total ideal is

$$\begin{aligned} \mathfrak{M} &= [\mathfrak{S}_1, \mathfrak{S}_2] = \mathfrak{S}_1 \mathfrak{S}_2 \\ &= ((x-1)(y-1)x^3, (y-1)x^2y, (y-1)xy^2, (x-1)z, y(y-1)x^3, yz), \end{aligned}$$

where

$$\begin{aligned} 1 &= -(y-1)x^3 + (y-1)(x^3-1) + y, \\ (y-1)(x^3-1) + y &\equiv 0 \pmod{\mathfrak{S}_1}, \quad -(y-1)x^3 \equiv 0 \pmod{\mathfrak{S}_2}. \end{aligned}$$

Here the ideals $\mathfrak{B}_1, \mathfrak{B}_2, \mathfrak{B}_3$ are isolated, and hence are uniquely determined also in the decompositions into irreducible and greatest primary ideals. The ideals $\mathfrak{B}_4$ and $\mathfrak{B}_5$, respectively $\mathfrak{Q}_4$, are non-isolated; they are not uniquely determined, but can, for instance, be replaced by

$$\mathfrak{D}_4 = (x^3, y + \lambda x^2, z), \quad \mathfrak{D}_5 = (x^2 + \mu xy, y^2, z),$$

and $\mathfrak{Q}_4$ can likewise be replaced by

$$\bar{\mathfrak{Q}}_4 = (x^3, x^2y, y^2 + \lambda xy, z).$$

2. Just as the general, and the integral, polynomial domains satisfy the finiteness condition, so does every finite integral domain of polynomials, by Hilbert's theorem on module bases;⁴⁶ the coefficients may be assigned to an arbitrary field. We give one more example, for the domain of all even polynomials, the simplest domain in which, because $x^2y^2 = (xy)^2$, no unique product representation of the polynomials by irreducible polynomials of the domain exists. It is the same configuration as in the preceding example, now given by the irreducible ideals

$$\begin{aligned} \mathfrak{B}_1 &= (x^2 - 1, xy, y^2, yz), \quad \mathfrak{B}_2 = (y^2 - 1, xz, yz, z^2), \\ \mathfrak{B}_3 &= (x^2, xy, xz, yz, z^2), \quad \mathfrak{B}_4 = (x^4, x^3y, y^2, xz, yz, z^2), \\ \mathfrak{B}_5 &= (x^2, y^2, xz, yz, z^2). \end{aligned}$$

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x^2, xy, y^2, xz, yz, z^2).$$

That $\mathfrak{P}_1$ is a prime ideal follows from the fact that every polynomial of the domain has the form

$$f = \varphi(z^2) + xz\psi(z^2) \pmod{\mathfrak{P}_1}.$$

Thus if

$$f_1 = \varphi_1(z^2) + xz\psi_1(z^2), \quad f_2 = \varphi_2(z^2) + xz\psi_2(z^2),$$

then from $f_1 f_2 \equiv 0 \pmod{\mathfrak{P}_1}$ one obtains the equations

$$\varphi_1 \varphi_2 + z^2 \psi_1 \psi_2 = 0, \quad \varphi_2 \psi_1 + \varphi_1 \psi_2 = 0,$$

⁴⁶Conversely, if a polynomial domain satisfies the finiteness condition and every polynomial has at least one representation in which the multipliers are of lower degree in x , then the domain is a finite integral domain.

and hence $f_1 \equiv 0 (\mathfrak{P}_1)$ or $f_2 \equiv 0 (\mathfrak{P}_1)$. In exactly the same way one sees that $\mathfrak{P}_2$ is a prime ideal; $\mathfrak{P}_3$ is a prime ideal because every polynomial of the domain modulo $\mathfrak{P}_3$ is congruent to a polynomial in y^2 ; $\mathfrak{P}_4$ consists of all polynomials of the domain. Thus $\mathfrak{B}_1$, $\mathfrak{B}_2$, and $\mathfrak{B}_3$ are irreducible as prime ideals, while $\mathfrak{B}_4$ and $\mathfrak{B}_5$ each have only the single proper divisor $\mathfrak{P}_4$, and hence are necessarily irreducible.

From the irreducible ideals one obtains the greatest primary ideals

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^4, x^3y, y^2, xz, yz, z^2),$$

the relatively-prime irreducible ideals

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^4, x^3y, x^2y^2, xy^3, xz, yz, z^2),$$

and the coprime-irreducible ideals

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y^2 - 1)x^4, (y^2 - 1)x^3y, (y^2 - 1)x^2y^2, (y^2 - 1)xy^3, xz, yz, z^2).$$

As in the first example,

$$[\mathfrak{B}_4, \mathfrak{B}_5] = [\mathfrak{D}_4, \mathfrak{D}_5], \quad [\mathfrak{Q}_3, \mathfrak{Q}_4] = [\mathfrak{Q}_3, \overline{\mathfrak{Q}}_4],$$

where

$$\mathfrak{D}_4 = (x^4, xy + \lambda x^2, \dots), \quad \mathfrak{D}_5 = (x^2 + \mu xy, \dots), \quad \lambda\mu \neq 1,$$

and

$$\overline{\mathfrak{Q}}_4 = (x^4, x^3y, y^2 + \lambda xy, \dots).$$

The remaining irreducible, respectively greatest primary, ideals $\mathfrak{B}_1$, $\mathfrak{B}_2$, $\mathfrak{B}_3$ are uniquely determined as isolated ideals.

§ 11. Examples from number theory and from the theory of differential expressions.

1. Let the underlying domain Σ consist of all even rational integers. Thus Σ can be put in one-to-one correspondence with the domain of all rational integers, by assigning to each number $2a$ in Σ the number a . It follows at once that every ideal in Σ is a principal ideal $(2a)$, where in the basis representation $2c = n \cdot 2a$ for every element $2c$ of the ideal, the odd integers n are only abbreviations for finite sums.

The prime ideals of the domain are $\mathfrak{P}_0 = \mathfrak{D} = (2)$ and $\mathfrak{P} = (2p)$, where p denotes an odd prime number; every prime ideal is therefore divisible by $\mathfrak{P}_0$, but by no further prime ideal. The primary ideals are

$$\mathfrak{Q}_0 = (2 \cdot 2^{e_0}) \quad \text{and} \quad \mathfrak{Q}_p = (2p^e).$$

They are at the same time irreducible ideals, and by what was said about the prime ideals, two ideals belonging to different odd prime numbers are mutually relatively prime, but no $\mathfrak{Q}$ is relatively prime to any $\mathfrak{Q}_0$.⁴⁷ Thus the $\mathfrak{Q}_0$ are the only non-isolated primary ideals. The unique decomposition of a into prime powers corresponds to the unique representation of the ideal $(2a)$ by greatest primary, at the same time irreducible, ideals:

$$(2a) = [(2 \cdot 2^{e_0}), (2p_1^{e_1}), \dots, (2p_\mu^{e_\mu})].$$

Thus here, in contrast to the examples from the polynomial domain, even the non-isolated greatest primary ideals are uniquely determined. If $e_0 = 0$ this is at the same time a representation by mutually relatively prime ideals, while if $e_0 > 0$ the ideal is relatively-prime-irreducible.

Thus, while in the domain of all integers the four different decompositions coincide, here this happens only for the two decompositions into greatest primary and into irreducible ideals on the one hand, and, for $e_0 > 0$, for coprime-irreducible (every ideal is coprime-irreducible, since the domain has no unit) and relatively-prime-irreducible ideals on the other; for $e_0 = 0$, however, the coprime-irreducible and relatively-prime-irreducible decompositions become different. At the same time one already obtains here an example in which one prime ideal can be divisible by another without being identical with it; more generally, one sees that product representation does not follow from divisibility. The latter fact - a consequence of the absence of a unit in the domain - is also why no unique product representation of the numbers of the domain by irreducible numbers of the domain exists, even though every ideal is a principal ideal; the introduction of least common multiples

⁴⁷Indeed, from $2b \cdot 2p_1^{e_1} \equiv 0 (2p_2^{e_2})$ for odd $p_1 \neq p_2$ one always obtains $2b \equiv 0 (2p_2^{e_2})$; but from $2b \cdot 2p^e \equiv 0 (2 \cdot 2^{e_0})$ one obtains only $2b = 2 \cdot 2^{e_0-1} \pmod{2 \cdot 2^{e_0}}$.

therefore proves necessary. It should also be noted that the situation remains exactly the same if, instead of all even integers, one takes all integers divisible by a fixed prime number or power of a prime.

Irreducible and primary ideals also become different, however, if Σ consists of all numbers divisible by a composite number $g = p_1^{\nu_1} \cdots p_s^{\nu_s}$. Again every ideal is a principal ideal $(g \cdot a)$, and the prime ideals are again $\mathfrak{P}_0 = \mathfrak{D} = (g)$ and $\mathfrak{P} = (g \cdot p)$, where p is a prime different from the primes occurring in g . But the irreducible ideals are $\mathfrak{B}_{\lambda_i} = (g \cdot p_i^{\lambda_i})$ and $\mathfrak{B}_e = (g \cdot p^e)$; the primary ideals are $\mathfrak{Q}_e = \mathfrak{B}_e$ and the ideals, different from the irreducible ones,

$$\mathfrak{Q}_{\lambda_1 \dots \lambda_s} = (g \cdot p_1^{\lambda_1} \cdots p_s^{\lambda_s}),$$

where the $\mathfrak{B}_{\lambda_i}$ and $\mathfrak{Q}_{\lambda_1 \dots \lambda_s}$ all have the same associated prime ideal $\mathfrak{P}_0 = (g)$. Here too the decomposition into irreducible ideals is unique, and consequently so is the decomposition into greatest primary ideals; thus here again the non-isolated ideals are uniquely determined.

2. An example of a non-commutative ring domain is supplied by the ideal theory in non-commutative polynomial domains treated in the paper of Noether–Schmeidler. In particular one has “completely reducible” ideals, that is, ideals for which the components of the decomposition are pairwise coprime and have no proper divisors; the components are therefore a fortiori irreducible. Thus, in addition to the isomorphism proved there according to § 9, one obtains equality of the number of components in two different decompositions. For the decomposition of systems of partial or ordinary linear differential expressions, which arises as a special case of that paper, this gives a result which does not seem to have been noticed even in the known case of a single ordinary linear differential expression.

At the same time the system T of all residue classes of a fixed ideal M , together with the non-commutative polynomial domain Σ , supplies a double domain (Σ, T) , where T has the module property with respect to Σ . For the difference of two residue classes is again a residue class, and likewise the product of a residue class by an arbitrary polynomial; whereas the product of two residue classes does not exist (loc. cit., § 3). Thus the systems of residue classes called “subgroups” there furnish examples of modules in double domains (Σ, T) , where the underlying ring domain Σ is non-commutative.

§ 12. Example from elementary divisor theory.

This is a conception of elementary divisor theory dictated by the general developments, but the theory itself is assumed as known.

Let Σ be the domain of all integral matrices with n^2 elements, with addition and multiplication defined in the usual sense for matrices. Then Σ is a non-commutative ring domain; the ideals are therefore in general one-sided, with two-sided ideals only as a special case.⁴⁸

We first show that every ideal is principal. For this purpose, in the case of right ideals, to each matrix $A = (a_{ik})$ we assign the module

$$A = (a_{11}\xi_1 + \cdots + a_{1n}\xi_n, \dots, a_{n1}\xi_1 + \cdots + a_{nn}\xi_n)$$

of integral linear forms. Conversely, to this module there corresponds every matrix that supplies a basis of A , hence, beside A , also UA , where U is unimodular. More generally, to the product PA there corresponds a module B which is a multiple of A . A single linear form from A is given by such matrices P as contain only one non-zero row.

Let now $A_1, A_2, \dots, A_\nu, \dots$ be all elements of an ideal $\mathfrak{M}$; let $A_1, A_2, \dots, A_\nu, \dots$ be the modules assigned to them; let A be their greatest common divisor and UA the most general matrix assigned to this module A . To each single linear form in A there then corresponds, by the definition of greatest common divisor, a matrix $P_1 A_{i_1} + \cdots + P_\sigma A_{i_\sigma}$, where, as above, the P have only one non-zero row. It follows that the matrix A corresponding to a basis of A also admits such a representation, now with general P , and hence becomes an element of $\mathfrak{M}$. Since, furthermore, every module A_i is divisible by A , every matrix A_i is divisible by A ; A , and in general UA , forms a basis of $\mathfrak{M}$. If one is dealing with left ideals, then correspondingly the columns of each matrix are to be regarded as the basis of a module; every ideal is a principal ideal, for which, beside A , also AV is a basis, with V an arbitrary unimodular matrix.

For the connection with elementary divisor theory, we now base the following on two-sided ideals; hence in particular PAQ belongs to the ideal whenever A does. By the preceding, the most general basis of such an ideal is UAV , where U and V are unimodular. The basis elements therefore exhaust a class of equivalent

⁴⁸The ideal theory of these domains is the subject of the works of Du Pasquier: *Zahlentheorie der Tettarionen*, dissertation, Zürich, Vierteljahrsschr. d. Naturf. Ges. Zürich, 51 (1906); *Zur Theorie der Tettarionenideale*, *ibid.*, 52 (1907). The content of the second paper is the proof that every ideal is a principal ideal.

matrices,⁴⁹ and there is a one-to-one relation between ideal and class, hence also between ideal and the elementary divisor system $(a_1 \mid a_2 \mid \cdots \mid a_n)$ of the class, where the a_i are, as is well known, non-negative integers, each dividing the following one. Thus the matrix of the class that occurs in the normal form determined by the elementary divisors may be regarded as a special basis of the ideal; divisibility of ideals, respectively of classes, implies divisibility of the elementary divisors, and conversely.

But Du Pasquier has shown, loc. cit. § 11, that for every two-sided ideal the rank is n and all elementary divisors agree. To be able to consider the case of general elementary divisors, we must therefore start not from the ideals but directly from the two-sided classes (which are likewise to be denoted by capital German letters). A class $\mathfrak{A} = UAV$ is divisible by another class $\mathfrak{B}$ if $A = PBQ$.

In general: *the least common multiple (the greatest common divisor) of two classes is obtained by forming the least common multiple (the greatest common divisor) of the corresponding elementary divisor systems.*⁵⁰ For let

$$(a_1 \mid a_2 \mid \cdots \mid a_n), \quad (b_1 \mid b_2 \mid \cdots \mid b_n), \quad (c_1 \mid c_2 \mid \cdots \mid c_n),$$

where $c_i = [a_i, b_i]$, be respectively the elementary divisor systems of $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}^*$, and let $\mathfrak{C} = [\mathfrak{A}, \mathfrak{B}]$. Then $\mathfrak{C}^*$ is divisible by $\mathfrak{A}$ and $\mathfrak{B}$, hence by $\mathfrak{C}$; conversely, the elementary divisors of $\mathfrak{C}$ are divisible by those of $\mathfrak{C}^*$, so $\mathfrak{C}$ is divisible by $\mathfrak{C}^*$ and therefore $\mathfrak{C} = \mathfrak{C}^*$. The proof for the greatest common divisor is obtained correspondingly.

The unique representation of the elementary divisors a_i as least common multiples of prime powers therefore corresponds to a representation of $\mathfrak{A}$ as the least common multiple of classes $\mathfrak{Q}$ whose elementary divisors are given by powers of one prime number, in symbols

$$\mathfrak{Q} \sim (p^{r_1} \mid p^{r_2} \mid \cdots \mid p^{r_\rho} \mid 0 \mid \cdots \mid 0), \quad r_1 \leq r_2 \leq \cdots \leq r_\rho.$$

If in particular the rank ρ is equal to n , then one has here a decomposition into coprime and coprime-irreducible classes which is unique despite the non-commutative domain.

The classes $\mathfrak{Q}$ can be further decomposed into classes corresponding to the elementary divisor systems

$$\begin{aligned} \mathfrak{B}_1 &\sim (p^{r_1} \mid \cdots \mid p^{r_1}), & \mathfrak{B}_2 &\sim (1 \mid p^{r_2} \mid \cdots \mid p^{r_2}), & \dots, \\ \mathfrak{B}_\nu &\sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu}), & \mathfrak{B}_{\rho+1} &\sim (1 \mid \cdots \mid 1 \mid 0 \mid \cdots \mid 0), \end{aligned}$$

where in $\mathfrak{B}_\nu$ the number 1 occurs $(\nu - 1)$ times. If, for example, $r_1 = r_2 = \cdots = r_\mu = 0$, then $\mathfrak{B}_1, \dots, \mathfrak{B}_\mu$ are equal to the unit class and are to be omitted from the decomposition; the same holds for $\mathfrak{B}_{\rho+1}$ if $\rho = n$. If, furthermore, say $r_\nu = r_{\nu+1} = \cdots = r_{\nu+\lambda}$, then $\mathfrak{B}_{\nu+1}, \dots, \mathfrak{B}_{\nu+\lambda}$ are proper divisors of $\mathfrak{B}_\nu$, and are likewise to be discarded. Denote by $\mathfrak{B}_{i_1}, \dots, \mathfrak{B}_{i_k}$ those that remain and now give a shortest representation. Then

$$\mathfrak{Q} = [\mathfrak{B}_{i_1}, \dots, \mathfrak{B}_{i_k}]$$

is the unique decomposition of $\mathfrak{Q}$ into irreducible classes. Indeed, suppose

$$\mathfrak{B}_\nu \sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu})$$

can be represented as the least common multiple of

$$\mathfrak{C} \sim (1 \mid \cdots \mid 1 \mid p^{s_\nu} \mid \cdots \mid p^{s_i}) \quad \text{and} \quad \mathfrak{D} \sim (1 \mid \cdots \mid 1 \mid p^{t_\nu} \mid \cdots \mid p^{t_i}).$$

Then in the elementary divisor systems of $\mathfrak{C}$ and $\mathfrak{D}$ the number 1 must stand in the first $(\nu - 1)$ places; in the ν -th place one of the exponents s_ν or t_ν , say s_ν , must become equal to r_ν . But since

$$r_\nu = s_\nu \leq s_{\nu+1} \leq \cdots \leq s_i \leq r_\nu,$$

one obtains $\mathfrak{C} = \mathfrak{B}_\nu$, and $\mathfrak{B}_\nu$ is therefore irreducible.⁵¹ The same holds for $\mathfrak{B}_{\rho+1}$, where p is to be replaced by 0. But, as the formation of the least common multiple shows, each of these irreducible classes gives

⁴⁹The basis elements of one-sided ideals correspond to right or left classes.

⁵⁰In the paper Zur Theorie der Moduln, Math. Ann. 52 (1899), p. 1, E. Steinitz defines the least common multiple (the greatest common divisor) of classes by the least common multiples and greatest common divisors of the elementary systems. Independently, the least common multiple of classes occurs as "congruence composition" in H. Brandt, Komposition der binären quadratischen Formen relativ einer Grundform, J. f. M. 150 (1919), p. 1.

⁵¹The $\mathfrak{B}$ are still reducible into one-sided classes; here there is no longer a unique relation between elementary divisors and class, and hence no unique decomposition into irreducible one-sided classes. The following example, supplied to me by H. Brandt, shows this for decomposition into right-sided classes (where the classes are represented by a basis matrix, or by the corresponding module):

$$\begin{aligned} \mathfrak{B} = [\mathfrak{C}_1, \mathfrak{C}_2] = [\mathfrak{D}_1, \mathfrak{D}_2], \quad \mathfrak{B} &\sim \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_1 \sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_2 \sim \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, \\ \mathfrak{D}_1 &\sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{D}_2 \sim \begin{pmatrix} p & 0 \\ p-1 & 1 \end{pmatrix}. \end{aligned}$$

Indeed, the modules $(\xi, p\eta)$, $(p\xi, \eta)$, and $(p\xi, (p-1)\xi + \eta)$ each have as their only proper divisor the module (ξ, η) , and are therefore irreducible and mutually distinct.

a definite exponent and the place where this exponent first occurs in the elementary divisor system of $\mathfrak{Q}$, respectively $\mathfrak{A}$, while $\mathfrak{B}_{\rho+1}$ gives the rank. Since these numbers are uniquely determined by the elementary divisor system of $\mathfrak{Q}$, respectively $\mathfrak{A}$, and since the relation between elementary divisors and class is one-to-one, the decomposition of $\mathfrak{Q}$, and likewise of any class, into irreducible classes is unique.

In summary: *Every two-sided class $\mathfrak{A}$ of integral matrices with bounded number of entries can be represented uniquely as the least common multiple of finitely many irreducible two-sided classes. Each irreducible class represents a fixed prime divisor of the elementary divisor system of $\mathfrak{A}$, an associated exponent, and the place where that exponent first occurs. The irreducible class corresponding to the divisor 0 gives the rank of $\mathfrak{A}$.*

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